Weekly Progress Report: SLM-to-SLM Guided Reasoning — Fine-Tuned Guidance Quality Over Model Size

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***Abstract*—This weekly report summarises the current progress of an ongoing research project investigating whether a fine-tuned Small Language Model (SLM) can replace large language model (LLM) guides in a guided reasoning pipeline. The pipeline pairs a LoRA fine-tuned Qwen2.5-3B guide with a Qwen2.5-1.5B solver ensemble using majority voting [2], evaluated across seven benchmarks. Key results confirm that fine-tuning quality—not model size—drives pipeline gains [6], with statistically confirmed accuracy improvements of +5.1 to +24.2 percentage points on four reasoning datasets.**

***Index Terms*—small language models, LoRA fine-tuning, guided reasoning, chain-of-thought, majority voting, position bias, calibration**

I. Project Overview

This research investigates whether a fine-tuned SLM can replace expensive LLM guides in a guided reasoning pipeline. A LoRA fine-tuned [6] Qwen2.5-3B guide generates structured verbal plans—drawing on chain-of-thought principles [1]—which are consumed by a Qwen2.5-1.5B solver ensemble using 5-vote majority voting [2]. The core hypothesis is that fine-tuning quality, not model size, is the key condition for effective SLM-to-SLM guidance, enabling strong reasoning performance at a fraction of the cost of LLM-based alternatives [3].

II. Completed Work This Week

A. Evaluation Across Seven Benchmarks

Full evaluation runs were completed across seven benchmarks spanning five domains: arithmetic (SVAMP N=300, ASDiv N=300), algebraic (AQUA-RAT N=100), science (ARC-Challenge N=100), commonsense (CommonsenseQA N=900), physical reasoning (PIQA N=500), and retrieval (RACE-High N=500). Results are summarised in Table I.

TABLE I
SUMMARY OF ACCURACY GAINS. ΔACC = GUIDED − BASELINE (PP). SIGNIFICANCE VIA MCNEMAR'S TEST [10].

Dataset	Base	Guided	ΔAcc	Sig.
PIQA	54.2%	78.4%	+24.2	$p{\approx}0$
SVAMP	40.3%	61.7%	+21.3	$p<.0001$
ASDiv	53.7%	64.0%	+10.3	$p=.0035$
CSQA	70.7%	75.8%	+5.1	$p<.001$
ARC-C	70.0%	77.0%	+7.0	ns
AQUA-RAT	40.0%	52.0%	+12.0	pending
RACE-High	58.6%	55.8%	−0.2	null

B. Position Bias Analysis

The fine-tuned guide substantially suppresses systematic position bias [8], [9] in the 1.5B solver. Most notably, A-option collapse in PIQA dropped from 93.4% to 42.1% (χ^2=54.34, p<0.001), accounting for a large fraction of the +24.2 pt accuracy gain. E-option preference in CommonsenseQA also decreased from 24.8% to 14.0%.

TABLE II
POSITION BIAS BEFORE AND AFTER GUIDANCE. LOWER % = LESS BIAS.

Dataset	Bias Type	Before	After
PIQA	Always picks A	93.4%	42.1%
CommonsenseQA	Prefers option E	24.8%	14.0%
ARC-Challenge	None detected	—	—

C. RACE-High Null Result

RACE-High at N=500 yields B=60, C=61, p=1.000—a clean null. The apparent +9 pt gain at N=100 was confirmed as sampling variance. This establishes the method's boundary condition: structured verbal planning helps tasks requiring *solution generation*, not *evidence retrieval*.

III. Calibration Findings

The expected calibration error (ECE) improved in all seven datasets, including RACE-High where the precision was null (sign test p=0.0078). ECE reduction ranges from 19.8% (CommonsenseQA) to 69.1% (PIQA), averaging 42.7% across the four confirmed datasets. This universality suggests structured planning reduces positional entropy in the solver's vote distribution independent of accuracy gain, consistent with prior calibration work [5].

IV. Vote Consistency

Guidance consistently increases the fraction of correct votes per question across all reasoning tasks. The correct-

TABLE III
CALIBRATION ERROR (ECE) BEFORE AND AFTER GUIDANCE. LOWER ECE = BETTER CALIBRATED CONFIDENCE.

Dataset	ECE Before	ECE After	Improvement
PIQA	0.354	0.110	69.1%
AQUA-RAT	0.272	0.148	45.6%
ASDiv	0.183	0.100	45.2%
ARC-Challenge	0.190	0.128	32.6%
SVAMP	0.202	0.128	36.6%
CommonsenseQA	0.176	0.141	19.8%

vote rate lift ranges from $1.09\times$ on CommonsenseQA and ARC-Challenge to $1.90\times$ on SVAMP. On CommonsenseQA (N=900), the guided pipeline shifts high-agreement correct questions from 603 to 680, while all-wrong questions decrease from 225 to 215—moving questions from the wrong pole to the correct pole without introducing middle-ground uncertainty.

TABLE IV
VOTE CONSISTENCY LIFT PER DATASET. C.L. = FRACTION OF CORRECT VOTES GAINED RELATIVE TO BASELINE.

Dataset	Correct-Vote Lift	Meaning
SVAMP	$1.90\times$	Nearly double correct votes
PIQA	$1.43\times$	43% more correct votes
ASDiv	$1.18\times$	18% more correct votes
ARC-Challenge	$1.09\times$	9% more correct votes
CommonsenseQA	$1.09\times$	9% more correct votes

V. COMPUTATIONAL COST

The pipeline is designed to remain compute-efficient. Total cost is measured in parameter-passes: the unguided baseline runs $1.5\text{B} \times 5 = 7.5\text{B}$ param-passes, the guided pipeline adds one guide pass for $3.0\text{B} \times 1 + 1.5\text{B} \times 5 = 10.5\text{B}$ param-passes, and the natural ceiling of running the 3B model for all five votes would cost 15.0B param-passes. The guided pipeline incurs only a 40% overhead over the baseline while delivering $+15.2$ pt average accuracy gain, and remains 30% cheaper than the size-matched ceiling [3].

TABLE V
COMPUTE COST VS. ACCURACY. PARAM-PASSES = TOTAL MODEL PARAMETERS $\times$ FORWARD PASSES.

Setup	Param-Passes	Avg Acc.	vs. Baseline
Baseline (1.5B $\times$ 5)	7.5B	54.7%	—
Guided (3B+1.5B$\times$5)	**10.5B**	**70.0%**	**+15.2 pt**
Ceiling (3B $\times$ 5)	15.0B	— (not tested yet)	—

VI. CONCLUSION

The core empirical finding is confirmed: fine-tuning quality, not model size, is the sufficient condition for effective SLM-to-SLM guidance. The pipeline achieves meaningful accuracy and calibration improvements at low additional compute cost. Key remaining work focuses on CoT and solo-3B baselines and completing the paper draft.

VII. REMAINING TASKS

The following items are pending:

- **CoT Baseline:** Add zero-shot chain-of-thought [4] baseline at equal compute (7.5B param-passes) for ASDiv and CommonsenseQA.
- **3B Solo Baseline:** Run Qwen2.5-3B (no guide, 5 votes) as a size-matched ceiling comparison.
- **SVAMP Ablation Re-run:** Re-run at solver temperature 0.4 to resolve the temperature confound.
- **AQUA-RAT Completion:** Finalise statistical testing and add McNemar results.
- **Error Analysis:** Systematic categorisation of guided-wrong/baseline-correct question types.
- **Paper Writing:** Begin drafting Method, Results, and Analysis sections.

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